Beyond Compensation: Persistent Income Gains from Land Expropriation in Rural China

Abstract

This paper evaluates the income effects of land expropriation on rural households in China. Drawing on the China Household Finance Survey (CHFS) and a staggered difference-in-differences (DID) framework, the analysis shows that households experience substantial income gains after land expropriation in both short and long run. This result is driven by the sizable lump-sum compensation and especially the subsequent reallocation of compensation into financial assets and adjustments in employment structure, which together sustain persistent income gains. I further find that the income effects do not vary significantly by household characteristics, but are more pronounced for households located closer to township centers. This paper provides new empirical evidence on the mechanisms and dynamics of income gains from land expropriation in developing countries.

Keywords: Land Expropriation, Property Rights, Farmer Income
JEL Codes: O18, Q15, R52

1. Introduction

In many developing countries, agriculture continues to employ a disproportionately large share of the labor force, yet the sector suffers from low labor productivity and incomes that lag behind those in urban and non-agricultural activities[1] (Schultz, 1951; Gollin et al., 2014). The overconcentration of labor force in agriculture further depresses value added per capita (Foster and Rosenzweig, 2022), reinforcing the rural-urban income gap. Consequently, how to raise rural household income has long been a

[1]See https://ourworldindata.org/employment-in-agriculture.

central question in developing economies.[2]

Governments pursue multiple policy toolkits to tackle this challenge. Common approaches include direct cash transfers (Parker and Todd, 2017), measures that broaden access to financial markets (Kaboski and Townsend, 2012), and investments that facilitate labor reallocation (Donaldson, 2018; Asher and Novosad, 2020).[3] Land policies constitute another important but less well-understood policy instrument within this toolkit (Adamopoulos et al., 2024, 2022; Besley and Ghatak, 2010). In particular, government-led land expropriation can, in principle, interact with these channels: it may provide lump-sum cash to households and encourage labor mobility. Yet despite their importance, land expropriation deprives farmers of land as a production factor. The persistent income consequences of such interventions remain poorly understood.

In this paper, I use the Chinese context to assess the extent to which Chinese government-led land expropriation raises rural household income. Using representative survey data from the China Household Finance Survey (CHFS) and a staggered Difference-in-Differences (DiD) design, I find land expropriation significantly increases household per capita income on average. Moreover, event study result reveals the distinct short and persistent dynamics effect: household income rises sharply within one to two years following expropriation, after which the magnitude declines but remains significantly positive.

I conduct a series of validation tests to assess the key assumptions underlying the staggered DiD framework, including the parallel trend assumption, no anticipation effects, and the Stable Unit Treatment Value Assumption (SUTVA). I also examine robustness to heterogeneous treatment effects. Since land expropriation is typically mandated by government and thus plausibly exogenous to individual households, I empirically compare key household characteristics between treatment and control groups, as well as examine whether treatment timing is correlated with household characteristics. The results show no significant differences between groups, and the timing of ex-

[2]According to China's official statistics, the share of agricultural income in rural household earnings fell from 71.4% in 1995 to 34.63% in 2022, while the share of wage income increased from 22.40% to 41.96%.

[3]Baird et al. (2011) compare conditional and unconditional cash transfer programs in Malawi, documenting effects on schooling and welfare. Burgess and Pande (2005) study the expansion of rural bank branches in India and find significant reductions in poverty.

propriation appears exogenous. Second, I test for potential spillover effects. I find that untreated households within treated villages experience a small and statistically insignificant income change, suggesting the absence of meaningful spillovers. Excluding such households does not materially change the baseline result. Finally, I assess treatment effect heterogeneity. Event study result remains consistent with two-way fixed effect (TWFE) regressions, suggesting that heterogeneous treatment effects are unlikely to bias my main findings.

In the mechanism analysis, I further decompose the dynamic treatment effects to illuminate the channels through which land expropriation affects household income. The evidence suggests that short-run income gains are predominantly driven by the sizable lump-sum compensation, which has grown considerably in recent years following institutional reforms (Huang et al., 2024). Over the longer horizon, two channels dominate. First, households allocate part of the compensation toward financial assets, generating additional returns that support persistent income gains. Second, the loss of land as a binding production factor reduces labor-mobility frictions, allowing household members to shift into higher-return non-agricultural activities.

In addition, I conduct heterogeneity analyses along household characteristics and village location. The income effects do not vary significantly across household demographic and socioeconomic characteristics. However, villages located closer to township centers experience substantially larger income gains, reflecting higher compensation and better access to non-agricultural employment opportunities.

This study contributes to two key areas of literature. First, it relates to the growing scholarship on property rights reforms. A large literature documents the profound economic consequences of China's property-rights transformations since 1978 (Lin, 1992; Chari et al., 2021; Adamopoulos et al., 2024). However, a specific gap exists concerning the effect of land expropriation on rural household income. Initial studies, grounded in the theory of insecure property rights, emphasized the negative consequences of expropriation risk for agricultural investment (Huntington and Shenoy, 2021; Jacoby et al., 2002; Besley and Ghatak, 2010) and farmers' subjective attitudes (Sha, 2023; Zhao and Xie, 2022). Over time, with the decline of the agricultural sector's economic contribution (Schultz, 1951; Gollin et al., 2014), the role of land expropriation in fostering income growth has become more apparent. Supporting this view, recent research indicates that land expropriation can positively affect individual health (Huang et al., 2024) and labor allocation

and migration (Zhang and Song, 2022; Ma and Mu, 2020; Wang et al., 2020).

While Huang et al. (2024) document the effects of land expropriation on individual health and labor reallocation, the persistent income dynamics and underlying financial mechanisms remain insufficiently understood. This paper contributes by providing new systematic evidence on long-run income growth net of compensation payments, documenting a complete financial channel from asset reallocation to measurable returns, and identifying meaningful spatial heterogeneity in treatment effects that has not been previously examined.

Second, the paper contributes to the broader literature on rural income dynamics in developing countries. A substantial body of work has examined policies designed to raise rural incomes, including targeted cash transfers (Baird et al., 2011; Parker and Todd, 2017), financial inclusion (Kaboski and Townsend, 2012; Burgess and Pande, 2005), and labor reallocation and mobility (Balboni et al., 2022; Gethin, 2025; Asher and Novosad, 2020; Donaldson, 2018). My findings suggest that land expropriation generates persistent income gains that extend well beyond the initial compensation payments, operating through asset accumulation and occupational reallocation from agriculture to non-agricultural activities. This complements the existing development literature by highlighting land policy as an underappreciated driver of rural income growth that operates through household-level financial and labor market channels.

The remainder of this study is organized as follows. Section 2 provides institutional background on Chinese land expropriation. Section 3 describes the data and empirical strategy. Section 4 presents the main results and robustness checks. Section 5 explores the underlying mechanisms. Section 6 presents heterogeneity analysis. Section 7 concludes this study.

2. Land Expropriation in China

China's land system is characterized by a distinctive dual land ownership system, with urban land owned by the state and rural land collectively owned by village collectives.[4] Rural land is primarily used for agricultural production, and farmers have use rights but not ownership rights. Registered villagers are granted the right to use collectively owned land for agricultural

[4]Under China's collective land ownership system, rural land is owned by village collectives, which represent all registered members of the village.

purposes under a contractual tenure, typically lasting 30 years, as stipulated in agreements signed with the village committee.

However, the state holds constitutional authority to expropriate rural collective land for purposes deemed to be in the public interest. In practice, both central and local governments may first convert collectively owned rural land into state-owned land and subsequently allocate it for infrastructure investment, industrial parks, and urban expansion, enabling China's rapid urbanization and industrialization. A key fiscal driver is that local governments can sell the use rights of expropriated land to developers at prices well above the statutory compensation paid to farmers, generating revenues that have become a central pillar of local public finance (*land finance*) (Gyourko et al., 2022; Han and Kung, 2015). As a result, urban boundaries have continued to expand while the stock of rural collective land has progressively diminished. As shown in Fig. 1, the three lines—representing total rural land area, agricultural land area, and arable land area—all exhibit a sustained downward trend, with the cumulative area of rural land expropriation between 2004 and 2022 reaching approximately 7 million hectares.

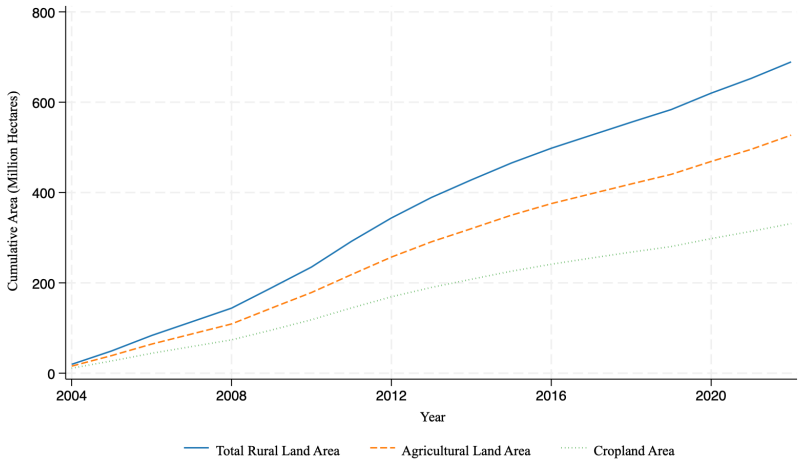


Fig. 1: Cumulative Expropriated Land Area in 2004-2022

Notes: The data are drawn from the China Land and Resources Statistical Yearbook and the China Natural Resources Statistical Yearbook. The three lines represent the total rural land area, agricultural land area, and arable land area, measured in 10,000 hectares.

This large-scale expropriation follows a standardized administrative pro-

cess governed by national legislation. When a local government plans to initiate land expropriation, it must first prepare a formal acquisition plan and submit it to higher-level authorities for review and approval. Upon receiving approval, the local government is required to publicly announce the expropriation plan to the villages and register landholdings for compensation assessment. The authorities then verify these registrations and prepare a detailed compensation and resettlement scheme. In the process of expropriation, there is no legal requirement for consultation with villagers before approval. Once compensation and resettlement have been implemented, the land is cleared and transferred for its designated use.[5]

Among the many issues surrounding land expropriation, compensation standards have been the most contentious. In practice, compensation is predominantly delivered in the form of lump-sum cash payment, sometimes supplemented by housing resettlement or social security enrollment. Under earlier legislation, the compensation standard was calculated as a multiple of the land's average annual agricultural output value—a formula that typically yielded amounts far below market value. Because many rural households relied heavily on agricultural land for their livelihoods, this below-market compensation often imposed substantial economic losses, leading to frequent disputes and conflicts in rural areas (Sha, 2023; Zhao and Xie, 2022).

Recent reforms have aimed to improve the land expropriation compensation and transparency, and strengthen the protection of farmers' rights. Table 1 summarizes major policy developments related to land expropriation, particularly the adjustments to compensation standards. For instance, the 2010 amendment to the *Land Administration Law* introduced stricter requirements for land expropriation, including the need for public hearings and greater equity in compensation. However, challenges remain: although households experiencing expropriation now receive sizable lump-sum cash compensation, the loss of land as a productive asset raises concerns about long-term livelihood sustainability once compensation is exhausted.

[5]The legal basis for this process is set out in Articles 46–50 of the *Land Administration Law of the People's Republic of China* (as amended in 2019) and further elaborated in the *Regulations for the Implementation of the Land Administration Law* (revised 2021).

Table 1: Major Reforms of Land Expropriation Compensation Standards in China

Year	Policy document	Main provisions
1982	*Measures for the Acquisition of Land for State Construction*	Land compensation: 3–6 times average annual output value (AAOV); resettlement subsidy: 2–3 times, capped at 10 times.
1998	*Land Administration Law*	Land compensation: 6–10 times AAOV; resettlement subsidy: 4–6 times; total 10–16 times, max 30 times in special cases.
2004	*Guiding Opinions on Improving Compensation and Resettlement*	Raised standards beyond 1998 law; emphasized livelihood protection.
2010	*Notice on Further Improving Land Expropriation Management*	Introduced "comprehensive land prices by zones" (3–5 zones); max up to 30 times AAOV; required direct, timely payments to farmers.

Note: This table outlines major national regulations defining land expropriation compensation standards in China. Earlier policies relied on AAOV-based multipliers, while the 2010 reform introduced "comprehensive land prices by zones," improving standardization and transparency. Compensation generally consists of land compensation and resettlement subsidies.

3. Data and Empirical Strategy

3.1. Data

This study draws on household- and individual-level panel data from the China Household Finance Survey (CHFS). The CHFS is a nationally representative survey administered by the Survey and Research Center for China Household Finance, designed to collect detailed micro-level information on household financial conditions (Gan et al., 2016). The survey covers 29 provinces in mainland China and is representative at both the national and provincial levels. It contains rich modules on economic activities, financial behavior, education and family structure. I use five waves (2013, 2015, 2017, 2019 and 2021) and match household- and individual-level variables across survey waves to construct a five-wave unbalanced panel dataset. I further merge village-level information from the CHFS village survey (available for 2013, 2015, 2017, and 2019), thereby obtaining a combined individual–household–village dataset.

Land expropriation is identified using the CHFS question that asks households to report the year in which their most recent land expropriation took

place.[6] At the village level, the CHFS collects a range of village-level variables, available for the 2013, 2015, 2017, and 2019 waves. Since time-invariant village characteristics are missing for certain observations, I impute them using within-village means across available waves, exploiting the panel structure of the data and the assumption that these characteristics remain stable over the survey period. More importantly, the CHFS attempts to re-interview households across waves while also expanding its sample over time, resulting in an unbalanced panel structure that nonetheless allows for the longitudinal tracking needed to assess the dynamic impact of land expropriation.

The analysis focuses on rural households with a household head aged above 16. I exclude households reporting negative income as they likely reflect survey measurement error and are incompatible with log transformation. Following existing studies (Huang et al., 2024; Sha, 2023), I also exclude households that report having experienced land expropriation before 2013. The final regression sample consists of 43,453 households across 652 villages. The share of households reporting land expropriation in the past two years is 15.56%, 14.97%, 15.34%, 14.48%, and 13.69% in 2013, 2015, 2017, 2019, and 2021, respectively, which are comparable to the expropriation rates calculated from the CFPS in Huang et al. (2024).[7] Table 2 displays descriptive statistics for key village-, household- and individual-level variables.

Table 2 presents the descriptive statistics. Panel A reports key village-level financial indicators, including collective assets, debt, income, and expenditure; some of these variables contain missing observations. In addition, Panel A summarizes several time-invariant baseline characteristics of villages,[8] such as the distance to the nearest township center, village area,

[6]The household members were asked "In which year did the most recent land expropriation take place?" The CHFS records only the most recent expropriation year. I identify households that likely experienced multiple rounds of expropriation as those whose reported year differs across survey waves, and confirm that results are robust to excluding them.

[7]In the balanced panel constructed from the CFPS (another nationally representative survey data in China), more than 25% of rural households have experienced land expropriation, while the overall share in the full sample is 15.2% (Huang et al., 2024). In the non-panel structure of the CFPS, the proportion of newly expropriated households in each wave between 2010 and 2018 is 12.1%, 6.4%, 6.6%, 5.7%, and 6.9%, respectively (Sha, 2023).

[8]To address inconsistencies in survey responses across waves, I take advantage of the panel structure of the data. For continuous variables, I compute the within-household

Table 2: Descriptive Statistics

	Observations	Mean	SD
Panel A. Village Variables			
Asset	1,848	1755.14	4449.94
Debt	1,891	266.95	704.60
Income	1,991	311.79	1240.83
Expenditure	1,838	267.73	724.24
Village Population	2,252	2005.87	1550.60
Distance to Township (km)	2,310	7.25	6.58
Village Area (*mu*)	2,310	72427.53	267005.34
Daxing (Yes=1)	2,310	0.80	0.29
Panel B. Household Variables			
Household Size	43,453	3.66	1.84
Per Capita Income	43,453	12.01	15.32
Per Capita Consumption	43,453	12.42	13.57
Per Capita Asset	43,453	102.00	186.40
Per Capita Debt	34,067	8.53	22.70
Per Capita Saving	43,453	4.46	12.99
CCP (Yes=1)	43,453	0.17	0.38
Head's Gender (Male=1)	43,453	0.87	0.34
Head's Age	43,453	56.81	12.26
Head's Years of Schooling	43,453	7.06	3.58
Panel C. Individual Variables			
Work Status	110,325	0.72	0.45
Agriculture	110,325	0.35	0.48
Non-agriculture	110,325	0.37	0.48
Other Jobs	79,921	0.08	0.27
Work Out	93,498	0.12	0.20

Note: Data are drawn from the China Household Finance Survey (CHFS) for 2013, 2015, 2017, 2019 and 2021, village variables are not available for 2021. All economic variables at both the village and household levels are winsorized. The *mu* is is a traditional Chinese unit of land area, commonly used for agricultural land and standardized as 1/15 of a hectare. Years of schooling are coded as follows: 0 = no schooling or incomplete primary education, 6 = primary school, 9 = junior high school, 12 = senior high school/vocational/technical school, 16 = college or university, and 19 = graduate-level education or above. The unit of all economic variables is thousand *yuan*. Individual-level work variables are measured as dummy indicators.

and the presence of *Daxing*. The *Daxing* is a dummy equal to 1 if a single surname accounts for over 10% of the village population. It proxies for the presence of a strong kinship-based clan structure, which is theorized to increase social cohesion and a village's capacity for collective action, potentially creating greater friction in land expropriation negotiations (Martinez-Bravo et al., 2022).

Panel B presents the household variables, with 43,453 household-wave observations. The variables include household financial conditions (per capita income, expenditure, assets, debt, and saving) as well as household size, whether the household has a CCP member, and basic characteristics of the household head (gender, age, and years of schooling). All economic and financial variables at both the village and household levels are measured in thousand *yuan*. Panel C presents individual-level work information based on 110,325 observations, including work status, participation in agricultural or non-agricultural activities, holding additional jobs, and outmigration. All variables are measured as dummy indicators.

3.2. Empirical Strategy

Staggered Difference-in-Differences. In light of the staggered rollout of the expropriation at the household level, I employ a staggered DiD framework that exploits variation in the timing of expropriation to identify its causal effects on household outcomes. A key advantage of this design is that it allows for an intuitive examination of dynamic outcome patterns relative to the expropriation event. The specification takes the form:

$$y_{ict} = \alpha + \beta DiD_{ict} + X_{ict}\phi + X_{c,2013} \times t + \gamma_t + \delta_c + \varepsilon_{ict} \tag{1}$$

Where y_{ict} denotes the logarithm of per capita household income for household i in village c at year t. DiD_{ict} is the treatment indicator of interest, which is equal to 1 after the household has ever been expropriated and 0 otherwise. X_{ict} represents household control variables, including household size, household CCP, the household head's gender, age and its square, and years of schooling. The term $X_{c,2013} \times t$ controls for differential time trends across villages by interacting a set of baseline (2013) village characteristics—including distance to township center, the presence of *Daxing*, and village

mean across all available waves. For the indicator of *Daxing*, I adjust the value using the modal response across waves.

area—with a linear time index, allowing villages with different baseline attributes to follow different income trajectories. γ_t and δ_c capture calendar year and village fixed effects, respectively.[9] The error term ε_{ict} is clustered at the village level, allowing for correlation across individuals within a village. β is the coefficient of interest that captures the treatment effect of state expropriation on per capita household income.

Event Study. Using the staggered DID does not require explicit consideration of heterogeneous treatment effects; however, it cannot capture the dynamic trajectory of outcomes following treatment. By contrast, event study complements DiD by allowing us to trace out the evolution of treatment effects over time (Miller, 2023). I estimate the event study to capture the dynamic effects of land expropriation:

$$y_{ict} = \sum_{r \neq 0} \mu_r DiD_{ic,t+r} + X_{ict}\phi + X_{c,2013} \times t + \gamma_t + \delta_c + \varepsilon_{ict} \tag{2}$$

where the dependent variable and control variables are defined in the same way as in equation (1). The coefficients of interest are μ_r, which measure the effect of r periods before and after the land expropriation. For clarity and precision, I aggregate event time into two-year intervals, collapsing adjacent pre- and post-treatment years into bins (e.g., $[-2, -1] \mapsto -1$, $[-4, -3] \mapsto -2$, and so forth), with year 0 defined as the baseline and excluded from the regression. In this specification, I expect the pre-treatment coefficients to be statistically indistinguishable from zero, consistent with the parallel trend assumption. By contrast, the post-treatment coefficients capture the dynamic policy effects, tracing the evolution of household income following the land expropriation.

The key identifying assumptions of the staggered DiD are parallel trend between treatment and control groups and no anticipation effects. Since land

[9]The choice to control for village fixed effects rather than household fixed effects in the baseline is guided by two primary considerations. First, given my use of a five-wave unbalanced panel dataset, the inclusion of high-dimensional fixed effects at the household level could amplify measurement error and increase the likelihood of spurious findings (Jennings et al., 2024). Second, since land expropriation is a treatment that is largely exogenous at the household level, potential omitted variables are more likely to operate at higher aggregation levels, such as the village. This makes village-level controls more pertinent for mitigating selection bias stemming from unobserved time-invariant local characteristics. Results are robust to replacing village fixed effects with household fixed effects in the robustness section.

Table 3: Balance Test

	2013	2015	2017	2019
Household size	-0.16	0.05	0.05	-0.61**
Per capita income	0.08*	-0.97	-0.62	0.72
Per capita consumption	0.07	-0.21	-0.58	0.93
Per capita asset	-1.21***	-13.25	-2.62	2.43
Per capita debt	0.27	1.04*	0.58	0.38
Per capita saving	-0.97	-0.38	-3.15	-0.10
CCP (Yes=1)	-0.08	0.03	-0.03	-0.05
Head's age	-0.10	-0.73	-0.07	1.83
Head's years of schooling	-0.59	-0.57	-0.19	-1.15*
F-statistic	1.37	0.98	0.60	1.16
Num. of clusters	376	506	582	548

Note: This table shows the t-statistics across four-wave key household variables between treatment and control groups. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

expropriation is implemented according to local government planning and is largely mandatory, households are unlikely to influence its implementation or timing, making self-selection bias an unlikely concern (Huang et al., 2024; Ma and Mu, 2020; Sha, 2023).

A key strength of the identification strategy in my paper is that treatment and control households are drawn from the same village. Village fixed effects absorb all time-invariant village-level confounders, including geographic advantages, local development potential, and government planning priorities. In particular, this within-village comparison addresses the concern that income gains reflect broader regional development trends rather than the causal effect of land expropriation, as both groups face identical local economic conditions. The identifying variation therefore comes from within-village differences in expropriation status, which are plausibly orthogonal to household-level characteristics.

Nevertheless, I formally test these assumptions in the following sections. For each wave of expropriations (2013, 2015, 2017, and 2019), I compare households about to be treated with all not-yet-treated households using data from the prior year, calculating t-statistics for mean differences across key household variables. As shown in Table 3, I find no evidence of systematic

pre-treatment differences, alleviating concerns about household self-selection and strengthening the credibility of my identification strategy.

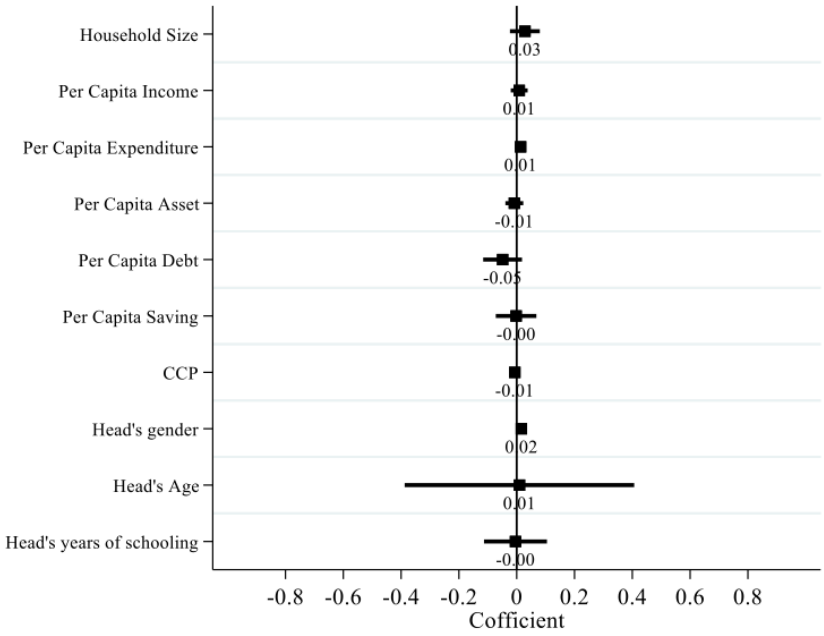


Fig. 2: Exogeneity of Treatment Timing

Notes: This figure shows conditional correlations between the timing of the land expropriation and pre-determined characteristics. I use several pre-determined characteristics as the dependent variable and where the independent variable is the expropriation year. The solid lines present the 90 percent confidence interval. I also control for calendar year and village fixed effects and all standard errors are clustered at the village level.

In the staggered DiD framework, it is necessary to further examine whether the timing of treatment is endogenous, i.e., whether the year of land expropriation is randomly assigned. To address this concern, I regress the expropriation year on the baseline household characteristics reported in Table 2, controlling for calendar year and village fixed effects, with standard errors clustered at the village level. The result presented in Fig. 2 shows no significant correlation between household baseline characteristics and the timing of land expropriation. This finding suggests that the timing of treatment is unlikely to be driven by household-specific factors, thereby supporting the

validity of my identification strategy.

4. Results

4.1. Main Results

Following land expropriation, it is reasonable to anticipate that households' income will increase due to compensation payments. Using the staggered DiD estimation in equation (1), I obtain the average treatment effect of land expropriation on household per capita income. The results are reported in Table 4. Column (1) presents the baseline estimate, showing that households affected by land expropriation experienced an average increase in per capita income of 31.6%. Column (2) adds household-level controls, yielding a coefficient of 29.2%. Given that village-specific characteristics may influence the income effects of land expropriation, in Column (3) I additionally include baseline characteristic $\times$ linear trend interactions, and the coefficient remains similar to Column (2).

Overall, the baseline results indicate that land expropriation significantly raises household per capita income, with an estimated increase of about 30%. Given that the mean per capita income in the control group is approximately 12,050 *yuan*, this corresponds to an average additional income of 3,615 *yuan* per person per year, or about 13,231 *yuan* per household (average household size is 3.66 shown in Table 2). While such compensation payments mechanically inflate household income in the short run, they come at the cost of losing land as a productive asset. The more fundamental question is whether households can sustain income growth in the long run once compensation is depleted—and if so, through what channels.

Next, following the equation (2), I plot an event study of household per capita income to examine the dynamic income changes of households subject to land expropriation. Fig. 3 illustrates the evolution of household income around the timing of land expropriation. The horizontal axis denotes years relative to land expropriation, with 0 indicating the year of expropriation. Each point in the figure represents the estimated coefficient of the corresponding time dummy, with the solid line indicating the 95 percent confidence interval. The event study result shows that, prior to land expropriation, there are no significant differences in income between the treatment and control groups, suggesting that the pre-treatment dynamics satisfy the parallel trend assumption.

Table 4: Basic Results

	(1)	(2)	(3)
	Log per capita income		
Land expropriation	0.315***	0.292***	0.293***
	(0.034)	(0.031)	(0.032)
Village FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Household controls	No	Yes	Yes
Baseline Char. × Trend	No	No	Yes
Mean of Dep. Var.	12.01	12.01	12.01
Obs.	43,453	43,453	43,453
Adj. R-squared	0.117	0.155	0.155

Note: This table shows the average effects of land expropriation on household per capita income. All regressions control for calendar year and village fixed effects. Column (2) adds household controls, including household size, household CCP, head's gender, age and its square, as well as years of schooling. In Column (3), I additionally control for the time-varying trends of baseline village characteristics, such as distance to the nearest township center, the presence of *Daxing* and village area. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

After land expropriation, household per capita income in the treatment group rises sharply, with the estimated effect exceeding 40%. This initial spike is likely driven by the lump-sum compensation payments, which are recorded as household income and generate a sudden, one-time boost. Importantly, the increase does not dissipate over time. Even 5–6 years after expropriation, household income remains significantly higher than in the pre-treatment period, indicating a persistent and long-lasting increase in household income. This dynamic pattern from the event study graph suggests that households initially benefit from lump-sum compensation within the first 1-2 years after land expropriation, while also experiencing sustained income growth in the longer run.

4.2. Robustness Checks

Households that experience land expropriation may generate spillover effects on other households within the same village that were not directly affected, potentially biasing the baseline estimates. To address this concern,

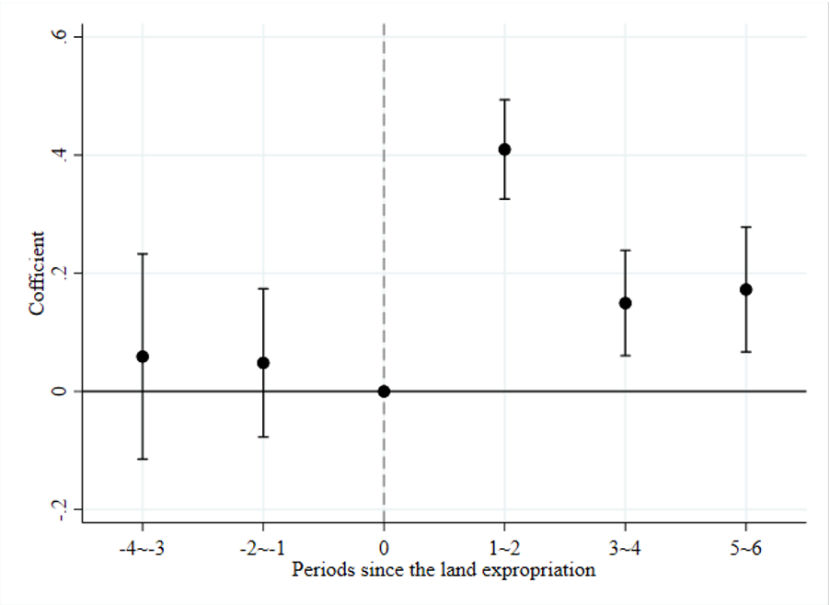


Fig. 3: Dynamic Effect of Household Income

Notes: This graph plots the coefficients and associated 95 percent confidence intervals from estimating the land expropriation on household per capita income. All the effects are relative to the two-year interval prior to the land expropriation. Data are drawn from the CHFS.

I exploit the fact that the CHFS provides both household- and village-level information: while households report whether and when they experienced land expropriation, village administrators also indicate whether the village as a whole was subject to expropriation. To reconcile potential discrepancies, I define a village as treated if at least one household in the village reports expropriation, even if the administrator does not confirm it. This approach allows me to construct treatment status at the village level using both sources of information.

In the baseline, the control group was defined as households that had not personally experienced land expropriation. In this section, I first redefine the control group as households residing in villages without any land expropriation, thereby excluding households in villages with expropriation but who themselves were not directly affected. This adjustment tests whether the baseline results are robust after removing potential spillover contamination.

Second, I explicitly test for spillover effects by setting the treatment group as households in villages where land expropriation occurred but were not directly expropriated, and the control group as households in villages without any expropriation.

Column (1) of Table 5 reports coefficients similar to the baseline findings, suggesting that excluding households in treated villages who were not directly expropriated does not materially affect the main results. Column (2) shows that households residing in treated villages but not directly affected by expropriation experience no significant income gains, with a small and statistically insignificant coefficient of 0.062. This finding indicates the absence of meaningful spillover effects on non-expropriated households within treated villages, supporting the validity of the Stable Unit Treatment Value Assumption (SUTVA) underlying the staggered DiD framework.

Table 5: Spillover Effects of Land Expropriation

	(1) Excluding potentially spillover group	(2) Excluding directly affected group
Land expropriation	0.262***	0.062
	(0.036)	(0.049)
Village FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
Baseline Char. × Trend	Yes	Yes
Obs.	35,250	39,535
Adj. R-squared	0.158	0.149

Note: All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Next, I conduct a series of robustness checks to validate the baseline findings. (1) I first control for county-level land titling reform. China's rural land titling program formalized farmers' contractual land rights through county-by-county certification campaigns actively promoted by the central

government from 2009 onwards.[10] Since land certification can independently affect household income by strengthening tenure security and facilitating land transfers (Chari et al., 2021), and since the reform rolled out at different times across counties—often overlapping with land expropriation events—failing to account for it could confound the estimated income effects. I manually collected the county-level roll-out year from official sources and constructed a dummy variable indicating whether certification had occurred prior to household expropriation.

I then conduct four additional checks. (2) I exclude households that experienced multiple rounds of land expropriation, identified as those whose reported expropriation year differs across survey waves, to ensure treatment timing is cleanly defined. (3) I exclude the 2021 wave because the COVID-19 pandemic may have affected both the likelihood of land expropriation and household income. (4) I cluster standard errors at the county level to account for coordination in land use planning within counties. (5) I add time-varying county-level controls—including GDP, value added of primary and secondary industries, total loan amount, and agricultural machinery—to further mitigate concerns that income gains reflect broader local development trends.

While village fixed effects are preferred over household fixed effects to mitigate measurement error amplification in short panels (Jennings et al., 2024), I acknowledge that unobserved household heterogeneity may plausibly affect both expropriation likelihood and income growth. I therefore provide household fixed effects estimates as a robustness check. Importantly, household fixed effects also absorb time-invariant within-village heterogeneity—such as a household's land location, proximity to roads, or development potential—thereby addressing the concern that within-village selection on unobserved land characteristics may confound the baseline estimates.

Fig. 4 displays the coefficient on land expropriation across all six specifications, with 95% confidence intervals. All estimates are positive and statistically significant, confirming the reliability of the main findings.

Additionally, I examine robustness to heterogeneous treatment effects us-

[10]The program was mandated through the *Opinions of the State Council on Guiding the Orderly Transfer of Rural Land Management Rights and Developing Appropriate-Scale Agriculture* (2014), following earlier pilot rollouts guided by Ministry of Agriculture circulars beginning in 2009. See https://www.gov.cn/zhengce/2014-11/20/content_2781544.htm.

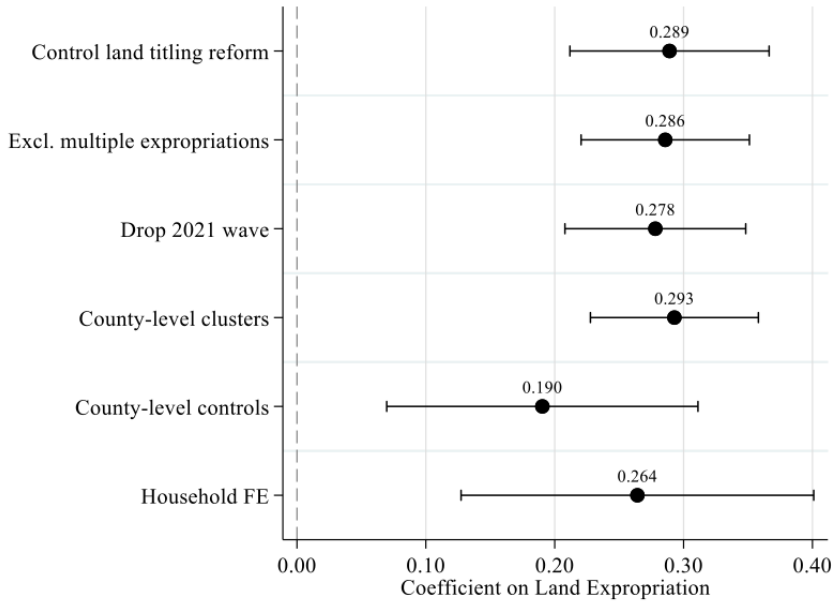


Fig. 4: Robustness Checks

Notes: This figure plots the coefficient on land expropriation from six alternative specifications, with 95% confidence intervals. (1) Controls for county-level land titling reform. (2) Excludes households with multiple expropriations. (3) Drops the 2021 survey wave. (4) Clusters standard errors at the county level. (5) Adds county-level time-varying controls (GDP, industrial value added, total loans, agricultural machinery). (6) Replaces village fixed effects with household fixed effects. All specifications include the same controls as the baseline regression. The dashed vertical line marks zero.

ing alternative staggered DiD estimators (Cengiz et al., 2019; Sun and Abraham, 2021). The corresponding results are presented in Fig. 5. Both the magnitude and statistical significance of the coefficients remain similar to the TWFE estimates, suggesting that my main findings are robust to heterogeneous treatment effects.

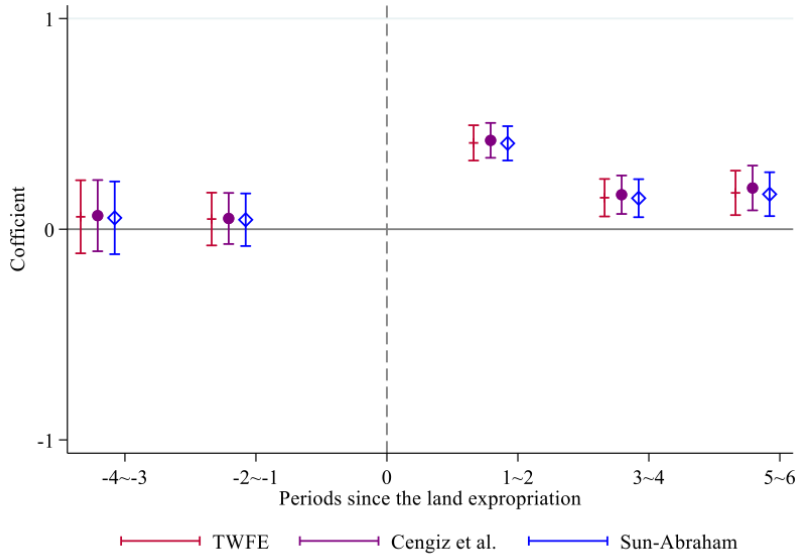


Fig. 5: Heterogeneous Treatment Effects

Notes: This figure overlays the event-study plots constructed using equation (2) with four different estimators: a dynamic version of the TWFE model, estimated using OLS (in cranberry with cross markers); Cengiz et al. (2019) (in purple with circle markers); and Sun and Abraham (2021) (in blue with diamond markers). The outcome variable is household per capita income. The horizontal axis denotes years relative to land expropriation, with 0 indicating the year of expropriation. To address the relatively small sample size in each individual year—which could otherwise lead to inflated variance and biased estimates—I aggregate the data into two-year intervals. Each point in the figure represents the estimated coefficient of the corresponding time dummy, with the solid line indicating the 95 percent confidence interval.

5. Mechanisms

5.1. Lump-sum Compensation in Cash

The first and most immediate channel through which land expropriation affects household income is the lump-sum cash compensation. With rising compensation standards in recent years, these transfers have become substantial. According to Huang et al. (2024), expropriated households received on average more than 6,000 *yuan* per capita immediately after land acquisition, equivalent to roughly 60% of their pre-expropriation income.

Drawing on CHFS data, I find that 77.85% of households in my sample received monetary compensation.[11] The average compensation across all households (including those who received nothing) was 31,270 *yuan*, with annual averages of 37,799 *yuan* in 2013, 27,591 *yuan* in 2015, 30,572 *yuan* in 2017, 28,647 *yuan* in 2019, and 31,740 *yuan* in 2021. For comparison, average annual household income in these years was 40,995 *yuan*, 45,962 *yuan*, 54,028 *yuan*, 53,358 *yuan*, and 54,062 *yuan*, respectively. A back-of-the-envelope calculation suggests that compensation accounted for about 62.94% of contemporaneous annual income, or 31.47% when spread over two years—figures broadly consistent with (Huang et al., 2024).

The baseline result shows that land expropriation raised household per capita income by an average of 30.0%, translating to approximately 13,217 *yuan* per household per year (0.30 $\times$ 3.66 members $\times$ 12,050). The event study further indicates that the cumulative income gain over a six-year horizon amounts to approximately 79,122 *yuan* per household, far exceeding the average one-time compensation of 31,270 *yuan*. This comparison suggests that lump-sum compensation, while substantial, cannot fully account for the long-run income gains identified in my estimates.

A central concern is that compensation payments are recorded as income in the CHFS, which could mechanically inflate the baseline estimate. To address this, I construct an alternative income measure excluding compensation receipts and re-estimate equation (1). Table 6 presents these results alongside the baseline. Column (2) shows that the income effect net of compensation remains positive and significant at 9.30%,[12] confirming that income gains

[11] Sha (2023) finds that 70.89% of households received compensation in cash, while 17.43% received no compensation. Other forms of compensation included household registration conversion, provision of job opportunities, and social insurance.

[12] I subtract compensation receipts from total household income, divide by household

reflect genuine behavioral responses beyond the mechanical compensation effect. This net-of-compensation result is the central finding of this paper: even after stripping out the one-time transfer, rural households experience a persistent and economically meaningful income gain of approximately 9% following land expropriation. In the following sections, I attribute this persistent growth to asset reallocation and adjustments in employment structure.

Table 6: Land Expropriation and Income Net of Compensation

	(1) ln(Income per capita)	(2) ln(Income net of compensation)
Land expropriation	0.293***	0.093***
	(0.032)	(0.032)
Controls	Yes	Yes
Baseline Char. × Trend	Yes	Yes
Village FE	Yes	Yes
Year FE	Yes	Yes
Obs.	43,453	43,278
Adj. R-squared	0.155	0.151

Note: All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Column (1) reports the baseline result. Column (2) constructs an alternative income measure excluding compensation receipts to isolate genuine behavioral responses beyond the direct compensation effect. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

5.2. Household Asset Allocation

The development of rural financial institutions has been shown to increase rural household income. Farmers expand income channels through increasing savings and investment (Prina, 2015), as well as adjusting agricultural investment and household entrepreneurship via access to credit (Banerjee and Newman, 1994). Government-led rural financial reforms have expanded access to formal financial institutions in rural areas, including village banks,

size to obtain per capita income net of compensation, replace missing values with zero, and winsorize at the 1st and 99th percentiles. The resulting coefficient of 9.30% is significant at the 1% level.

loan companies, and rural mutual cooperatives. For households experiencing land expropriation, the compensation provides a substantial lump-sum transfer, while the loss of land simultaneously depreciates their agricultural capital stock. As a result, these households are more likely to redirect funds toward financial institutions rather than agricultural investment (Huntington and Shenoy, 2021; Jacoby et al., 2002). In this section, I examine changes in household asset allocation following land expropriation, making use of the rich household financial variables provided by CHFS.

First, I focus on changes in household savings behavior. Although nearly half of rural households in my sample hold no savings accounts prior to expropriation, compensation receipts provide a strong incentive to open accounts and deposit funds. I separately estimate both the intensive margin (amount of savings per capita) and extensive margin (probability of holding savings) effects. Second, I examine changes in household assets, including per capita total assets, financial assets, and non-financial assets (agricultural and non-agricultural assets).[13]

Table 7 presents the results. Columns (1) and (2) show that both intensive and extensive margin of household savings per capita increased significantly after land expropriation, by 26.6% and 4.4 percentage points, respectively. Total household assets increased by 17.6% and financial assets by 31.5%. Among non-financial assets, agricultural assets declined (although the coefficient is statistically insignificant), while other non-agricultural assets increased significantly. These results suggest two main findings: (1) households tend to allocate compensation from land expropriation into savings and financial investments; (2) with the loss of land as a productive factor, households shift away from agricultural assets and reallocate investment toward financial assets and other non-agricultural assets, leading to a significant change in asset composition.

I further classify household assets by risk level and estimate effects on both the intensive and extensive margins (Table B.2 in the Appendix). The

[13]Specifically, based on CHFS definitions, total assets consist of financial and non-financial assets. Financial assets include cash, deposits, stocks, bonds, funds, financial products, derivatives, foreign-currency assets, gold, loans to others, and internet financial assets. Non-financial assets include agricultural assets (livestock, agricultural inventory, and machinery used for farming) and other non-agricultural assets (commercial and industrial assets, vehicles such as cars and tricycles, and durable goods such as computers, washing machines, and artworks).

Table 7: Asset Allocation

	(1) Saving	(2) Saving (0-1)	(3) Assets	(4) Fin. assets	(5) Agri. assets	(6) Non-fin. assets
Land expropriation	0.266***	0.044***	0.176***	0.315***	-0.027	0.111***
	(0.070)	(0.012)	(0.038)	(0.055)	(0.072)	(0.038)
Village FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Char. × Trend	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Var.	4.46	0.49	102.00	7.72	1.03	6.45
Obs.	21,267	43,453	43,453	39,790	22,935	43,453
Adj. R-squared	0.173	0.115	0.265	0.177	0.875	0.306

Notes: All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Column (1) represents the continuous variable of per capita household savings. Column (2) is a dummy variable indicating whether the household has savings. The remaining variables are tail-truncated and get the logarithm form. Total household per capita assets include financial assets and non-financial assets. Non-financial assets include agricultural and other non-financial assets. Agricultural assets include livestock, agricultural product inventory, and agricultural machinery (set to 0 if the household is not engaged in agriculture). Other non-financial assets mainly include business assets, vehicles (cars, motorcycles), and durable goods (computers, washing machines, books, etc.). The variation in observations across columns reflects the presence of zero values in some asset categories: columns (1), (4), and (5) apply a log transformation and thus drop households with zero savings, zero financial assets, and zero agricultural assets, respectively. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

results show that, following land expropriation, rural households exhibit significant increases in risky assets along both margins. In contrast, while holdings of risk-free assets increase on the intensive margin, no significant change is observed on the extensive margin. These findings further suggest that, after receiving sizable lump-sum compensation, households tend to reallocate their portfolios toward higher-risk and potentially higher-return assets. The event study in Fig. A.1 in the Appendix further confirms that household per capita assets increase substantially in the periods immediately following land expropriation and that this growth persists over the longer term, suggesting that the asset accumulation is not solely attributable to the one-time compensation payments but reflects sustained appreciation.

To provide more direct evidence on this channel, I directly estimate the effect of land expropriation on household asset returns. Table 8 shows that land expropriation significantly increases both the probability of receiving asset returns by 3.5 percentage points and the magnitude of returns by 24.7%, providing direct evidence that asset accumulation following expropriation generates sustained income gains rather than merely reflecting a mechanical increase in asset holdings.[14] This portfolio-level evidence, including the distinction between risky and risk-free assets and the direct estimation of asset returns, goes beyond what has been documented in existing work and provides a more complete account of the financial mechanisms underlying persistent income growth.

5.3. Employment Structure Adjustment

According to data from China's National Bureau of Statistics,[15] the share of agricultural employment fell from 70.5% in 1978 to 23.6% in 2020. Two major transformations account for this change: the increase in the percentage of households living and working in cities, and the shift in employment structure among households remaining in villages (Adamopoulos et al., 2024). Land expropriation serves as an important driver of this reallocation by reducing mobility friction and accelerating the shift of rural labor into non-agricultural activities (Ma and Mu, 2020; Zhang and Song, 2022), thereby sustaining income gains beyond the initial compensation.

[14] Asset returns are measured using a log(1+x) transformation to accommodate zero values in the full sample.

[15] See the China Statistical Yearbook at https://www.stats.gov.cn/sj/ndsj/.

Table 8: Asset Returns Following Land Expropriation

	(1) Asset return (dummy)	(2) Log asset return
Land expropriation	0.035***	0.247***
	(0.010)	(0.065)
Village FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
Baseline Char. × Trend	Yes	Yes
Obs.	43,453	43,453
Adj. R-squared	0.240	0.152

Note: All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Column (1) is a dummy variable indicating whether the household has asset returns. Column (2) uses a $\log(1+x)$ transformation of asset returns to accommodate zero values in the full sample. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Previous studies find that the transition of farmers into non-agricultural employment enhances household labor productivity and income (Ngai et al., 2019; Adamopoulos et al., 2024, 2022). Building on this literature, I examine how land expropriation induces persistent income gains through adjustments in employment structure. The analysis is conducted at the individual level, restricting the sample to working-age individuals aged 16–60. Specifically, I use each individual's primary occupation to construct the following outcomes: a dummy for current employment status; dummies for engagement in agricultural and non-agricultural work (the latter including wage employment, self-employment, and flexible employment); a dummy for holding a second job; and a dummy for working outside the village.

Table 9 reports the results. I find that: (1) The probability of being employed did not change significantly after land expropriation, suggesting that individuals did not withdraw from the labor market despite the household losing land. (2) The composition of employment shifted: the probability of engaging in agricultural work declined, while the probability of non-agricultural employment increased, though overall work intensity did not rise. (3) Unlike prior findings (Ma and Mu, 2020; Zhang and Song, 2022), I document a decline in the probability of working outside the village after land expro-

Table 9: Employment Structure and Allocation

	(1) Work status	(2) Agri. share	(3) Non-agri. share	(4) Second job	(5) Out-migration
Land expropriation	-0.005	-0.022**	0.017*	0.002	-0.009
	(0.008)	(0.010)	(0.010)	(0.007)	(0.007)
Village FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Baseline Char. × Trend	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Var.	0.72	0.35	0.37	0.08	0.12
Obs.	100,068	100,068	100,068	72,497	84,868
Adj. R-squared	0.041	0.118	0.069	0.042	0.130

Notes: All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Column (1) is a dummy variable indicating whether household members are working after land expropriation. Columns (2) and (3) show the share of household members engaged in agricultural and non-agricultural employment. Column (4) indicates whether household members have a second job. Column (5) uses the share of migrant workers in the household as the dependent variable, where migrants refer to those working outside, studying, or serving in the military. We restrict samples between 16 and 60 years old. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

priation, although the coefficient is statistically insignificant. In sum, Table 9 indicates that while land expropriation reshaped employment structure by encouraging a shift toward non-agricultural jobs within the village, it shows no significant effect on out-migration.

Importantly, the absence of significant changes in overall employment rates and second-job holding suggests that these gains reflect improvements in allocative efficiency—occupational switching from low-productivity agricultural work to higher-return non-agricultural activities—rather than an expansion of labor supply. The income gains documented in this paper therefore arise from where workers are employed, not from how much they work.

The magnitudes of the employment reallocation effects are economically meaningful relative to their baseline shares: the decline in agricultural employment (-0.022) amounts to roughly 6.3% of the baseline agricultural share (0.35), and the increase in non-agricultural employment (0.017) corresponds to about 4.6% of the baseline non-agricultural share (0.37). To further val-

idate that the significant shift into non-agricultural work in Table 9 reflects a genuine reallocation effect rather than a statistical artifact, I replicate the regressions restricting the sample to individuals aged above 60—past the typical retirement threshold—and report the results in Table B.3 in the Appendix. Among this older group, land expropriation is associated with a significant decline in agricultural employment (-0.034^{**}), but the corresponding shift into non-agricultural work is statistically indistinguishable from zero. This contrast has a clear interpretation: older individuals who lose agricultural land have no viable pathway into non-agricultural employment, consistent with well-documented age barriers in China's labor market. The fact that the non-agricultural effect appears only in the working-age sample but not among those above 60 supports the conclusion that land expropriation does induce meaningful agricultural-to-non-agricultural labor reallocation for those able to make the transition.

Given that land expropriation induces adjustments in labor allocation and thereby raises household income, a natural question arises: why did households not shift from agricultural to non-agricultural employment prior to expropriation? I argue that two factors may account for this result. First, land expropriation is to some extent compulsory and imposed by the state. Once households lose access to their land, they are forced to reallocate labor toward non-agricultural sectors. Second, China's incomplete land tenure system constrains labor mobility. Rural residents face implicit incentives to maintain active land cultivation, as households that have already exited agriculture may face uncertainty over compensation entitlements in future expropriation events. This discourages exit from agriculture and keeps some households in low-productivity farming, resulting in labor-mobility friction (Adamopoulos et al., 2024; Chen et al., 2022).

6. Heterogeneity

In this section, I investigate the heterogeneity in the income effects of land expropriation along two dimensions: household characteristics and township location.

6.1. Household Characteristics

Households with different demographic and socioeconomic attributes may exhibit distinct labor and asset allocation responses to land expropriation,

thereby influencing the income effects. I focus on the household head's education and age, as well as whether the household has a village cadre or CCP member. Previous studies have shown that these characteristics may directly or indirectly affect household employment or financial decisions. For example, Xie (2019) finds that older individuals and those with lower human capital experience more from land expropriation, while men are more likely to out-migrate.

In this subsection, I augment equation (1) with interaction terms between the treatment indicator and baseline household characteristics, implementing a triple difference (DDD) estimation. Education is defined as a binary indicator equal to 1 if the household head has completed high school or above. Working-age population is coded as 1 if the household head is between 18 and 65 years old, and gender is coded as 1 if male. Finally, I split households based on whether they have a cadre or CCP member. Table 10 reports the results. The DDD estimates are statistically insignificant across all specifications, and the point estimates do not reveal systematic patterns along any of the dimensions examined, suggesting that the income effects of land expropriation do not vary substantially by household characteristics.

6.2. Village Location

I next explore spatial heterogeneity with respect to distance to township center of household. Households located closer to township may experience larger income gains, both because their land is more likely to be expropriated at higher compensation rates and because they have more opportunities to adjust employment and asset structure. To test this, I partition the sample into distance bands of 0-5 km, 5-10 km, 10-15 km, and 15-20 km from township center and estimate the equation (1) for each subsample. Fig. 6 plots the results. The estimated effects are strongest within 5 km and decline with distance, becoming statistically insignificant beyond 10 km, consistent with significant spatial heterogeneity in treatment effects.

To further validate this finding, I calculate the compensation by distance based on CHFS. The average compensation across all households in 2013-2021 is 11,844 *yuan*. In contrast, households within 5 km of township received about 13,034 *yuan*. I also examine whether households closer to township center adjust their employment structure or asset allocation differently by distance. As shown in Fig. A.2 in the Appendix, household savings increase significantly in areas closer to township centers, while households within 15

Table 10: Heterogeneity Analysis: Household Characteristics

	(1)	(2)	(3)	(4)
	Log per capita income			
Land expropriation	0.296***	0.274***	0.292***	0.309***
	(0.033)	(0.054)	(0.035)	(0.037)
Land expropriation × Education	-0.019			
	(0.067)			
Land expropriation × Working-age popu.		0.026		
		(0.061)		
Land expropriation × Village cadre			-0.092	
			(0.103)	
Land expropriation × CPC member				-0.075
				(0.070)
Village FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Baseline Char. × Trend	Yes	Yes	Yes	Yes
Obs.	43,453	43,453	37,610	43,453
Adj. R-squared	0.155	0.155	0.162	0.155

Notes: This table presents the heterogeneous effect by household characteristics. All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. In Column (1), Education is defined as a dummy variable equal to 1 if the household head has high school or above. Column (2) defines working-age population as household head aged 18-65 years old. Columns (3) and (4) investigate differences between households with and without village cadre or CPC membership, respectively. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

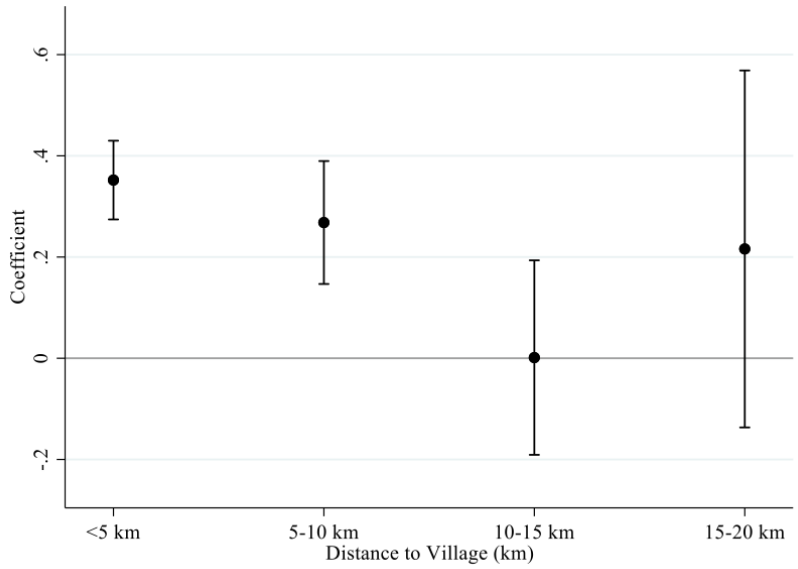


Fig. 6: Spatial Heterogeneity

Notes: Based on the survey question that defines the distance from the township center, I use this measure to capture spatial heterogeneity. I partition the sample into distance bands of 0-5 km, 5-10 km, 10-15 km, and 15-20 km from township center and estimate each subsample. The results are presented with 95 percent confidence intervals.

km are more likely to shift from agricultural to non-agricultural employment. These patterns confirm that both mechanism channels are stronger for households closer to township centers.

Taken together, these results suggest that the income effects of land expropriation are concentrated among households located closer to township centers, likely due to higher compensation and greater opportunities for labor and asset structural adjustment.

7. Conclusion

This paper provides new evidence on the income effects of land expropriation by analyzing rural households in China using micro-level survey data and a staggered Difference-in-Differences design. I find that land expropriation significantly raises household income, with distinct short- and long-term

dynamics. While short-run gains are largely driven by direct compensation payments, the central finding of this paper is that income gains persist even after stripping out the one-time compensation: households experience a persistent and economically meaningful income gain of approximately 9% beyond the direct compensation effect. This longer-run growth operates through two channels: reallocation of compensation into financial assets that generate measurable returns, and a shift in employment from agriculture to higher-return non-agricultural activities within the village. These findings underscore the role of land expropriation not only as a redistribution mechanism but also as a driver of household-level income growth through asset accumulation and labor reallocation.

The results also reveal meaningful spatial heterogeneity. Households located closer to township centers experience substantially larger income gains, reflecting both higher compensation standards and greater access to non-agricultural labor markets. This spatial gradient in treatment effects suggests that the proximity to urban economic activity is a key moderator of how land expropriation translates into household welfare improvements—a finding with direct implications for where and how expropriation programs are designed.

More broadly, the analysis highlights that in developing countries where agriculture still absorbs a large share of the labor force and labor productivity remains low, land-related institutional reforms can play a central role in raising rural household incomes. By easing frictions to labor mobility and providing the means for asset diversification, land expropriation—if accompanied by adequate and transparent compensation—can contribute to sustained rural income growth. These lessons extend beyond China and offer insights into how developing economies may design land and labor policies to balance equity, efficiency, and long-term development goals. Future research could further explore the distributional consequences of these income gains and whether the asset-accumulation channel persists as compensation standards and financial market access continue to evolve.

References

Adamopoulos, T., Brandt, L., Chen, C., Restuccia, D., Wei, X., 2024. Land security and mobility frictions. The Quarterly Journal of Economics 139, 1941–1987.

Adamopoulos, T., Brandt, L., Leight, J., Restuccia, D., 2022. Misallocation, selection, and productivity: A quantitative analysis with panel data from china. Econometrica 90, 1261–1282.

Asher, S., Novosad, P., 2020. Rural roads and local economic development. American Economic Review 110, 797–823.

Baird, S., McIntosh, C., Özler, B., 2011. Cash or condition? evidence from a cash transfer experiment. The Quarterly Journal of Economics 126, 1709–1753.

Balboni, C., Bandiera, O., Burgess, R., Ghatak, M., Heil, A., 2022. Why do people stay poor? The Quarterly Journal of Economics 137, 785–844.

Banerjee, A.V., Newman, A.F., 1994. Poverty, incentives, and development. The American Economic Review 84, 211–215.

Besley, T., Ghatak, M., 2010. Property rights and economic development, in: Handbook of Development Economics. Elsevier. volume 5, pp. 4525–4595.

Burgess, R., Pande, R., 2005. Do rural banks matter? evidence from the indian social banking experiment. American Economic Review 95, 780–795.

Cengiz, D., Dube, A., Lindner, A., Zipperer, B., 2019. The effect of minimum wages on low-wage jobs. The Quarterly Journal of Economics 134, 1405–1454.

Chari, A., Liu, E.M., Wang, S.Y., Wang, Y., 2021. Property rights, land misallocation, and agricultural efficiency in china. The Review of Economic Studies 88, 1831–1862.

Chen, C., Restuccia, D., Santaeulàlia-Llopis, R., 2022. The effects of land markets on resource allocation and agricultural productivity. Review of Economic Dynamics 45, 41–54.

Donaldson, D., 2018. Railroads of the raj: Estimating the impact of transportation infrastructure. American Economic Review 108, 899–934.

Foster, A.D., Rosenzweig, M.R., 2022. Are there too many farms in the world? labor market transaction costs, machine capacities, and optimal farm size. Journal of Political Economy 130, 636–680.

Gan, L., Yin, Z., Tan, J., et al., 2016. Report on the development of household finance in rural China (2014). Springer.

Gethin, A., 2025. Distributional growth accounting: Education and the reduction of global poverty, 1980-2019. The Quarterly Journal of Economics , qjaf033.

Gollin, D., Lagakos, D., Waugh, M.E., 2014. The agricultural productivity gap. The Quarterly Journal of Economics 129, 939–993.

Gyourko, J., Shen, Y., Wu, J., Zhang, R., 2022. Land finance in china: Analysis and review. China Economic Review 76, 101868.

Han, L., Kung, J.K.S., 2015. Fiscal incentives and policy choices of local governments: Evidence from china. Journal of Development Economics 116, 89–104.

Huang, W., Luo, M., Ta, Y., Wang, B., 2024. Land expropriation, household behaviors, and health outcomes: Evidence from china. Journal of Development Economics 171, 103358.

Huntington, H., Shenoy, A., 2021. Does insecure land tenure deter investment? evidence from a randomized controlled trial. Journal of Development Economics 150, 102632.

Jacoby, H.G., Li, G., Rozelle, S., 2002. Hazards of expropriation: tenure insecurity and investment in rural china. American Economic Review 92, 1420–1447.

Jennings, J., Kim, J.M., Lee, J., Taylor, D., 2024. Measurement error, fixed effects, and false positives in accounting research. Review of Accounting Studies 29, 959–995.

Kaboski, J.P., Townsend, R.M., 2012. The impact of credit on village economies. American Economic Journal: Applied Economics 4, 98–133.

Lin, J.Y., 1992. Rural reforms and agricultural growth in china. The American Economic Review , 34–51.

Ma, S., Mu, R., 2020. Forced off the farm? farmers' labor allocation response to land requisition in china. World Development 132, 104980.

Martinez-Bravo, M., Padró i Miquel, G., Qian, N., Yao, Y., 2022. The rise and fall of local elections in china. American Economic Review 112, 2921–2958.

Miller, D.L., 2023. An introductory guide to event study models. Journal of Economic Perspectives 37, 203–230.

Ngai, L.R., Pissarides, C.A., Wang, J., 2019. China's mobility barriers and employment allocations. Journal of the European Economic Association 17, 1617–1653.

Parker, S.W., Todd, P.E., 2017. Conditional cash transfers: The case of progresa/oportunidades. Journal of Economic Literature 55, 866–915.

Prina, S., 2015. Banking the poor via savings accounts: Evidence from a field experiment. Journal of Development Economics 115, 16–31.

Schultz, T.W., 1951. The declining economic importance of agricultural land. The Economic Journal 61, 725–740.

Sha, W., 2023. The political impacts of land expropriation in china. Journal of Development Economics 160, 102985.

Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Journal of Econometrics 225, 175–199.

Wang, C., Akgüç, M., Liu, X., Tani, M., 2020. Expropriation with hukou change and labour market outcomes in china. China Economic Review 60, 101391.

Xie, Y., 2019. Land expropriation, shock to employment, and employment differentiation: Findings from land-lost farmers in nanjing, china. Land Use Policy 87, 104040.

Zhang, C., Song, Y., 2022. Road to the city: Impact of land expropriation on farmers' urban settlement intention in china. Land Use Policy 123, 106432.

Zhao, X., Xie, Y., 2022. The effect of land expropriation on local political trust in china. Land Use Policy 114, 105966.

Appendix A. Figures

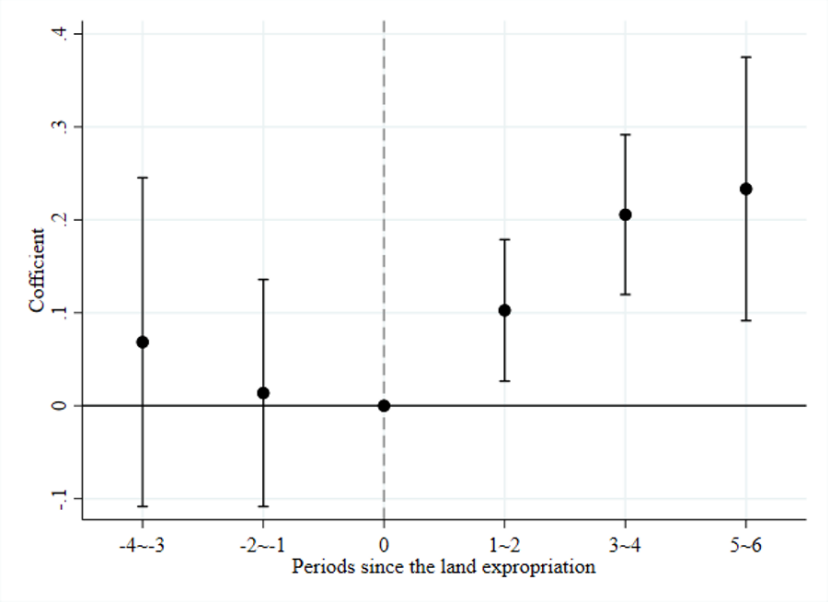


Fig. A.1: Dynamic Effect of Household Asset

Notes: The graphs plot the coefficients and associated 95 percent confidence intervals from estimating the land expropriation on household per capita assets. All the effects are relative to the two-year interval prior to the land expropriation. Data are obtained from the CHFS.

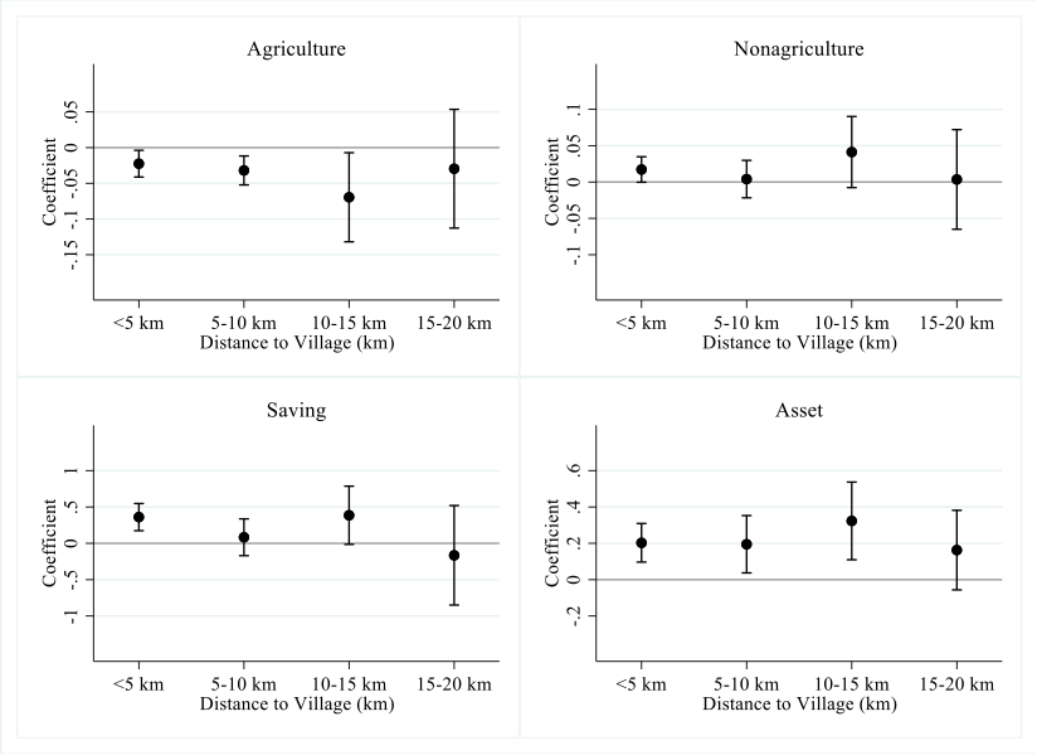


Fig. A.2: Spatial Heterogeneity of Mechanism

Notes: Following the same specification as in Fig. 6, I estimate key variables related to employment and household assets. Households within 15 km of township center exhibit significant shifts in employment structure and asset allocation, while those beyond 15 km do not.

Appendix B. Tables

Table B.1: Subjective Attitudes

	(1) Happiness	(2) Self-rated health	(3) Job service sati.	(4) Security
Land expropriation	-0.024	-0.020	-0.062	-0.071
	(0.024)	(0.022)	(0.040)	(0.048)
Village FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Baseline Char. $\times$ Trend	Yes	Yes	Yes	Yes
Obs.	43,757	43,817	17,627	11,145
Adj. R-squared	0.095	0.254	0.107	0.087

Notes: All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. All indicators are measured on a scale of 1-5, with higher scores indicating better conditions. The analysis includes questions about household happiness and self-rated health across all survey years; job service satisfaction in 2015, 2017, and 2019; and security indicators in 2015 and 2017. Due to these timing differences, some variables have missing observations. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table B.2: Financial Assets Classification

	Risky Financial Assets		Risk-free Financial Assets	
	Intensive	Extensive	Intensive	Extensive
Land Expropriation	0.254***	0.042***	0.282***	0.009
	(0.092)	(0.011)	(0.055)	(0.008)
Village FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Baseline Char. × Trend	Yes	Yes	Yes	Yes
Mean of Dep. Var.	5.70	0.36	1.73	0.89
Obs.	15,765	43,453	38,507	43,453
Adj. R-squared	0.170	0.491	0.191	0.061

Notes:All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Risky financial assets include stocks, bonds, funds, financial derivatives, financial products, non-RMB assets, gold, loans, and other financial assets; risk-free financial assets include cash, demand deposits, time deposits, and other bank account balances. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table B.3: Employment Structure Adjustment: Sample Aged Above 60

	(1) Agri. share	(2) Non-agri. share
Land expropriation	-0.034**	0.000
	(0.014)	(0.007)
Village FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
Baseline Char. × Trend	Yes	Yes
Obs.	41,787	41,787
Adj. R-squared	0.130	0.056

Notes: This table replicates the employment structure regressions from Table 9 restricting the individual-level sample to those aged above 60. All regressions control for calendar year and village fixed effects, as well as household-level controls and time-varying trends of baseline village characteristics. Standard errors are clustered at the village level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.